

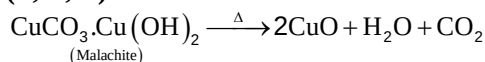
Chemistry

SECTION 1 (Maximum Marks: 12)

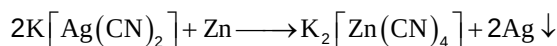
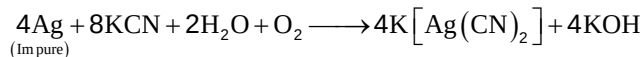
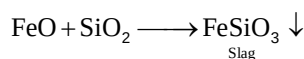
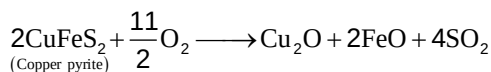
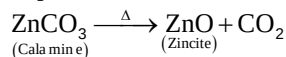
- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 choosing **ONLY** (A), (B) and (D) will get +4 marks;
 choosing **ONLY** (A) and (B) will get +2 marks;
 choosing **ONLY** (A) and (D) will get +2 marks;
 choosing **ONLY** (B) and (D) will get +2 marks;
 choosing **ONLY** (A) will get +1 mark;
 choosing **ONLY** (B) will get +1 mark;
 choosing **ONLY** (D) will get +1 mark;
 choosing no option (i.e. the question is unanswered) will get 0 marks; and
 choosing any other combination of options will get -2 marks.

- Q.1. The correct statement(s) related to processes involved in the extraction of metals is (are)
- (A) Roasting of Malachite produces Cuprite.
 (B) Calcination of Calamine produces Zincite.
 (C) Copper pyrites is heated with silica in a reverberatory furnace to remove iron.
 (D) Impure silver is treated with aqueous KCN in the presence of oxygen followed by reduction with zinc metal.

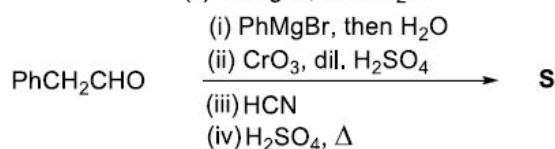
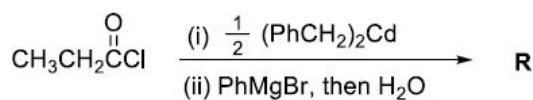
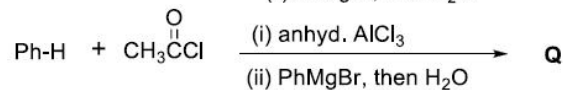
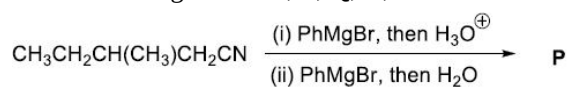
Sol. (B, C, D)



Cuprite – Cu_2O



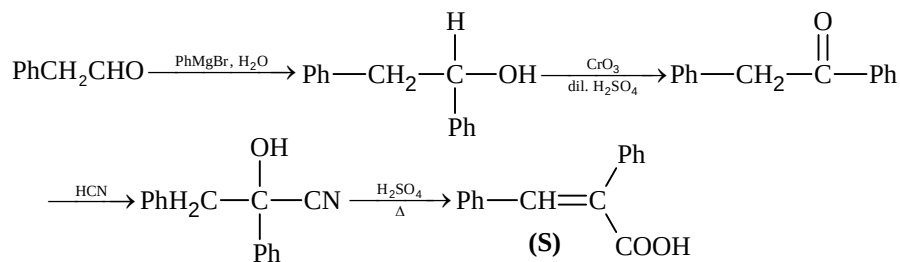
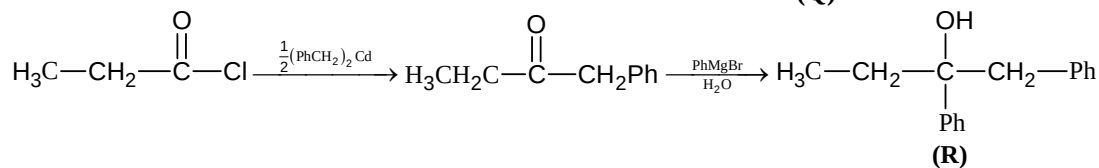
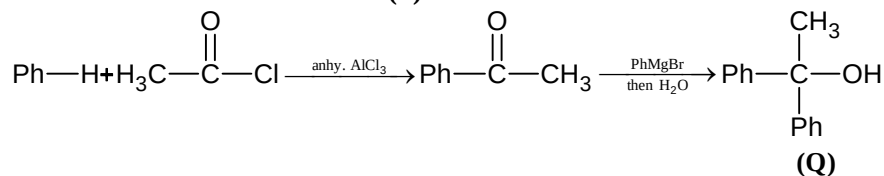
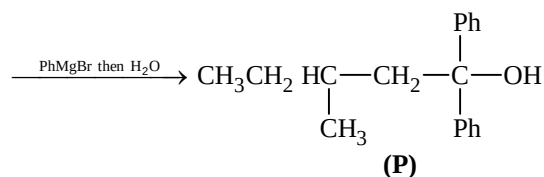
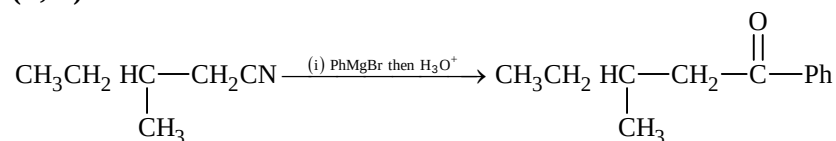
Q.2. In the following reactions, **P**, **Q**, **R**, and **S** are the major products.



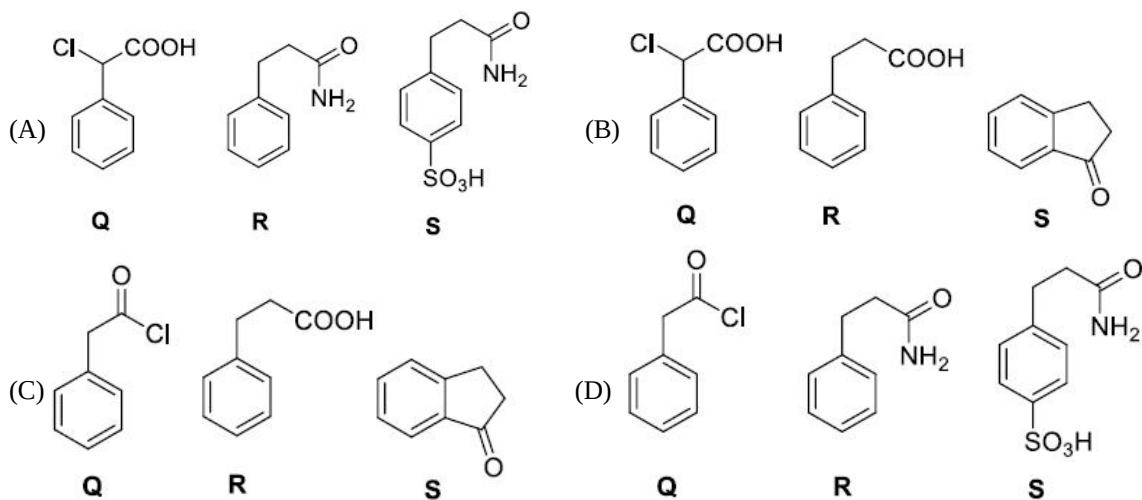
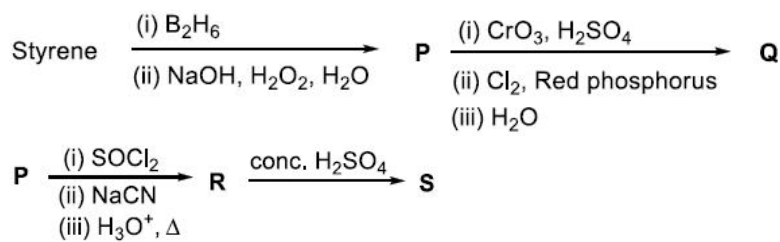
The correct statement(s) about **P**, **Q**, **R**, and **S** is (are)

- (A) Both **P** and **Q** have asymmetric carbon(s).
 (B) Both **Q** and **R** have asymmetric carbon(s).
 (C) Both **P** and **R** have asymmetric carbon(s).
 (D) **P** has asymmetric carbon(s), **S** does **not** have any asymmetric carbon.

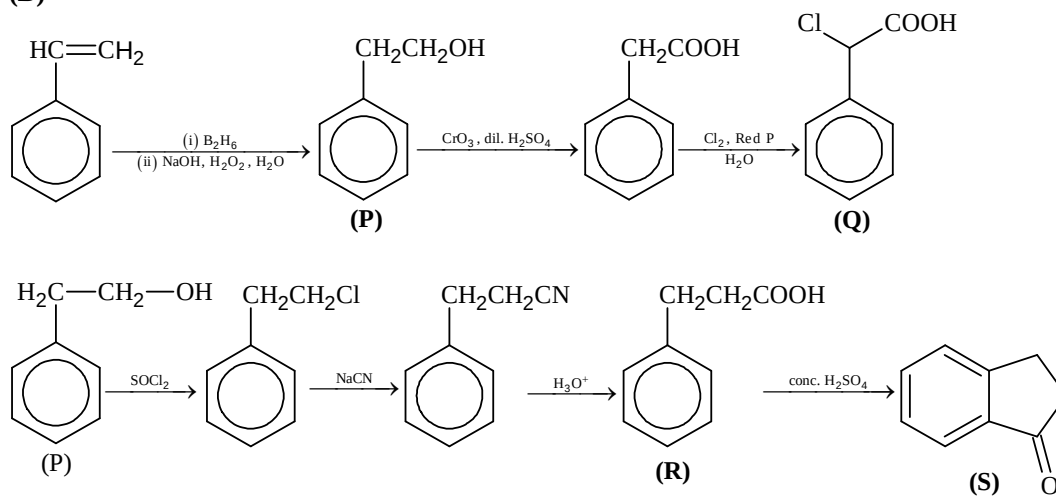
Sol. (C, D)



Q.3 Consider the following reaction scheme and choose the correct option(s) for the major products **Q**, **R** and **S**.



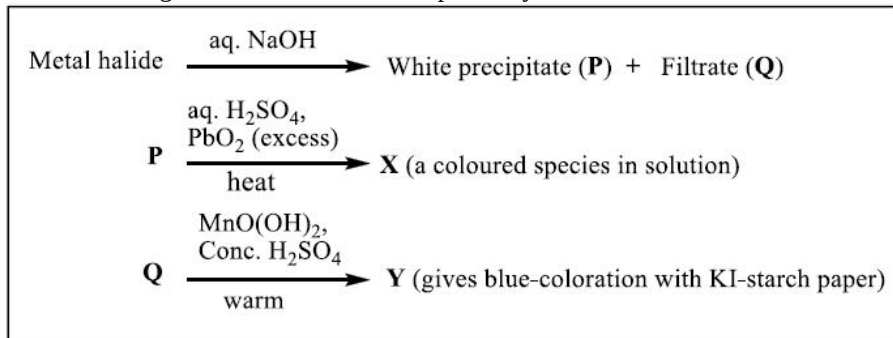
Sol. (B)



SECTION 2 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

Q.4 In the scheme given below, **X** and **Y**, respectively, are



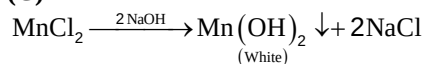
(A) CrO_4^{2-} and Br_2

(B) MnO_4^{2-} and Cl_2

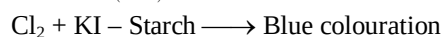
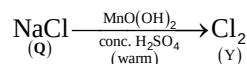
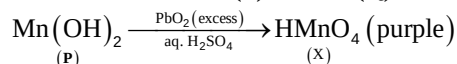
(C) MnO_4^- and Cl_2

(D) MnSO_4 and HOCl

Sol. (C)



(P) (Q)



Q.5 Plotting $1/\Lambda_m$ against $c\Lambda_m$ for aqueous solutions of a monobasic weak acid (HX) resulted in a straight line with y-axis intercept of P and slope of S. The ratio P/S is

$[\Lambda_m = \text{molar conductivity}]$

$\Lambda_m^0 = \text{limiting molar conductivity}$

$c = \text{molar concentration}$

$K_a = \text{dissociation constant of HX}$

(A) $K_a \Lambda_m^0$

(B) $K_a \Lambda_m^0 / 2$

(C) $2K_a \Lambda_m^0$

(D) $1 / (K_a \Lambda_m^0)$

Sol. (A)

$$K_a = \frac{c \left(\frac{\lambda_m}{\lambda_m^0} \right)^2}{\left(1 - \frac{\lambda_m}{\lambda_m^0} \right)}$$

$$K_a = \frac{C \frac{\lambda_m^2}{\lambda_m^0}}{\lambda_m^0 - \lambda_m}$$

$$K_a = \frac{C \lambda_m^2}{\lambda_m^0 (\lambda_m^0 - \lambda_m)}$$

$$K_a (\lambda_m^0)^2 - K_a \lambda_m \lambda_m^0 = C \lambda_m^2$$

$$K_a \frac{(\lambda_m^0)^2}{\lambda_m} - K_a \lambda_m^0 = C \lambda_m$$

$$\frac{1}{\lambda_m} = + \left(\frac{C \lambda_m}{K_a (\lambda_m^0)^2} \right) + \frac{K_a \lambda_m^0}{K_a (\lambda_m^0)^2}$$

$$\frac{1}{\lambda_m} = \frac{C \lambda_m}{K_a (\lambda_m^0)^2} + \frac{1}{\lambda_m^0}$$

$$S = \frac{1}{K_a (\lambda_m^0)^2}; \quad P = \frac{1}{\lambda_m^0}$$

$$\frac{P}{S} = K_a \lambda_m^0$$

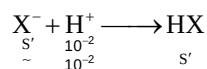
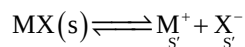
- *Q.6 On decreasing the pH from 7 to 2, the solubility of a sparingly soluble salt (MX) of a weak acid (HX) increased from $10^{-4} \text{ mol L}^{-1}$ to $10^{-3} \text{ mol L}^{-1}$. The $\text{p}K_a$ of HX is
- (A) 3 (B) 4
(C) 5 (D) 2

Sol. (B)

Lets assume solubility of the salt MX is S at pH = 7.



Lets assume solubility of the salt MX is S' at pH = 2.



$$\frac{[\text{HX}]}{[\text{X}^-][\text{H}^+]} = \frac{1}{K_a}$$

$$\frac{S'}{[\text{X}^-] 10^{-2}} = \frac{1}{K_a}$$

$$\frac{S'^2}{10^{-2}} = \frac{K_{sp}}{K_a} \quad \dots(2)$$

Equation (2) divided by equation (1)

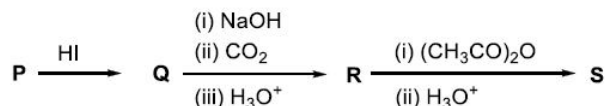
$$\frac{S'^2}{S^2 \times 10^{-2}} = \frac{K_{sp}}{K_a} \times \frac{1}{K_{sp}}$$

$$\frac{(10^{-3})^2}{(10^{-4})^2 \times 10^{-2}} = \frac{1}{K_a}$$

$$K_a = 10^{-4}$$

$$pK_{a(HX)} = 4$$

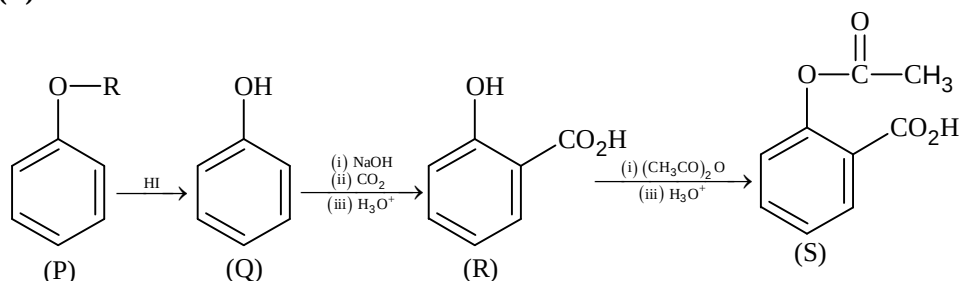
- Q.7 In the given reaction scheme, **P** is a phenyl alkyl ether, **Q** is an aromatic compound; **R** and **S** are the major products.



The correct statement about **S** is

- (A) It primarily inhibits noradrenaline degrading enzymes.
 (B) It inhibits the synthesis of prostaglandin.
 (C) It is a narcotic drug.
 (D) It is *ortho*-acetylbenzoic acid.

Sol. (B)



Aspirin is a non – narcotic analgesics.
 It inhibits the synthesis of prostaglandin.

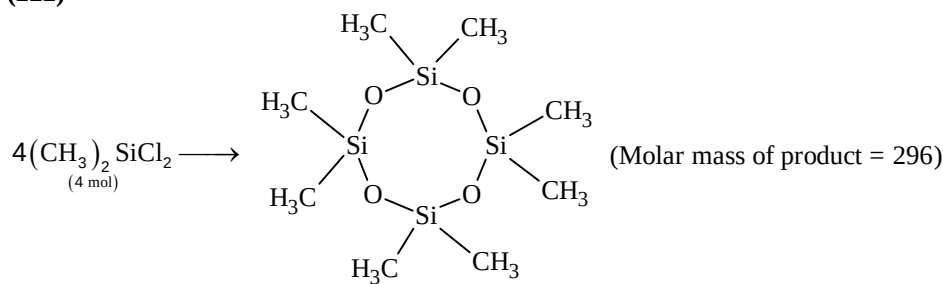
SECTION 3 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

- *Q.8 The stoichiometric reaction of 516 g of dimethyldichlorosilane with water results in a tetrameric cyclic product **X** in 75% yield. The weight (in g) of **X** obtained is ____.
 [Use, molar mass (g mol⁻¹): H = 1, C = 12, O = 16, Si = 28, Cl = 35.5]

Sol. (222)



4 mol $\longrightarrow$ 0.75 mol (because efficiency = 75 %)
 Mass of product = 0.75 $\times$ 296 = 222 gm

- *Q.9 A gas has a compressibility factor of 0.5 and a molar volume of $0.4 \text{ dm}^3 \text{ mol}^{-1}$ at a temperature of 800 K and pressure $x \text{ atm}$. If it shows ideal gas behaviour at the same temperature and pressure, the molar volume will be $y \text{ dm}^3 \text{ mol}^{-1}$. The value of x/y is ____.
[Use: Gas constant, $R = 8 \times 10^{-2} \text{ L atm K}^{-1} \text{ mol}^{-1}$]

Sol.

(100)

$$Z = 0.5$$

$$V = 0.4 \text{ dm}^3 \text{ mol}^{-1}$$

$$T = 800 \text{ K}$$

$$P = x$$

$$Z = \frac{PV}{RT} = 0.5 = \frac{x \times 0.4}{0.08 \times 800}$$

$$x = 80$$

When $Z = 1$, Ideal condition molar volume $y \text{ dm}^3$

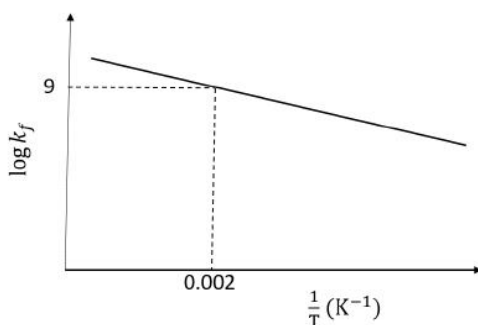
$$Z = \frac{PV}{RT}$$

$$1 = \frac{80 \times y}{0.08 \times 800}$$

$$y = 0.8$$

$$\frac{x}{y} = \frac{80}{0.8} = 100$$

- *Q.10 The plot of $\log k_f$ versus $1/T$ for a reversible reaction $A(g) \rightleftharpoons P(g)$ is shown



Pre-exponential factors for the forward and backward reactions are 10^{15} s^{-1} and 10^{11} s^{-1} , respectively. If the value of $\log K$ for the reaction at 500 K is 6, the value of $|\log k_b|$ at 250 K is ____.

[K = equilibrium constant of the reaction]

k_f = rate constant of forward reaction

k_b = rate constant of backward reaction]

Sol.

(5)

$$\log k_f = 9 \text{ at } 500 \text{ K}, \therefore k_f = 10^9$$

$$\log k_b = \frac{-(E_a)_b}{2.303R} \frac{1}{T} + \log A_b$$

$$K_{eq} = 10^6 = \frac{10^9}{k_b}$$

$$k_b = 10^3$$

$$3 = \frac{-(E_a)_b}{2.303R} \times 0.02 + 11$$

$$\frac{(E_a)_b}{2.303R} = \frac{8}{0.002} = 4000$$

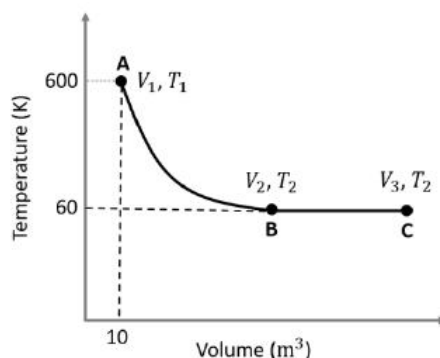
$$\log k_b = -4000 \times \frac{1}{250} + 11$$

$$= -16 + 11$$

$$= -5$$

$$|\log k_b| = 5$$

- *Q.11 One mole of an ideal monoatomic gas undergoes two reversible processes ($A \rightarrow B$ and $B \rightarrow C$) as shown in the given figure:



$A \rightarrow B$ is an adiabatic process. If the total heat absorbed in the entire process ($A \rightarrow B$ and $B \rightarrow C$) is $RT_2 \ln 10$, the value of $2 \log V_3$ is ____.

[Use, molar heat capacity of the gas at constant pressure, $C_{p,m} = \frac{5}{2}R$]

Sol. (7)

$A \rightarrow B$ (reversible adiabatic), $\gamma = \frac{5}{3}$

$$q_{\text{total}} = E_{AB} + q_{BC} = 0 + q_{BC} = q_{BC}$$

$$q_{BC} = RT_2 \ln 10$$

For the process A to B

$$T_1 \times V_1^{\gamma-1} = T_2 \times V_2^{\gamma-1}$$

$$\left(\frac{V_2}{V_1}\right)^{\gamma-1} = \frac{T_1}{T_2} = \frac{600}{60} = 10$$

$$\left(\frac{V_2}{V_1}\right)^{\frac{5}{3}-1} = 10$$

$$\frac{V_2}{V_1} = (10)^{\frac{3}{2}}$$

$$V_2 = 10 \times (10)^{\frac{3}{2}} = (10)^{\frac{5}{2}}$$

$$V_2 = (10)^{\frac{5}{2}}$$

$$-RT_2 \ln 10 = -RT_2 \ln \frac{V_3}{V_2}$$

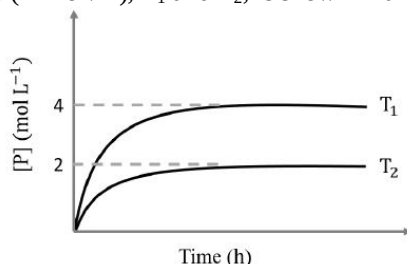
$$10 = \frac{V_3}{V_2}$$

$$V_3 = 10V_2 = 10 \times (10)^{\frac{5}{2}} = (10)^{\frac{7}{2}}$$

$$2 \ln V_3 = 2 \times \ln 10^{\frac{7}{2}}$$

$$2 \log V_3 = 2 \times \frac{7}{2} \times 1 = 7$$

- *Q.12 In a one-litre flask, 6 moles of A undergoes the reaction $A(g) \rightleftharpoons P(g)$. The progress of product formation at two temperatures (in Kelvin), T_1 and T_2 , is shown in the figure:



If $T_1 = 2T_2$ and $(\Delta G_2^\circ - \Delta G_1^\circ) = RT_2 \ln x$, then the value of x is

[[ΔG_1° and ΔG_2° are standard Gibb's free energy change for the reaction at temperatures T_1 and T_2 , respectively.]

Sol. (8)

	$A(g) \rightleftharpoons P(g)$	
Initial mole	6	0
at eq. at T_1	2	4
	$(K_{eq})_{T_1} = \frac{4}{2} = 2$	
at eq. T_2	4	2
	$(K_{eq})_{T_2} = \frac{2}{4} = \frac{1}{2}$	

$$\Delta G_1^\circ = -RT_1 \ln 2$$

$$\Delta G_2^\circ = -RT_2 \ln \frac{1}{2} = RT_2 \ln 2$$

$$\Delta G_2^\circ - \Delta G_1^\circ = RT_2 \ln 2 + RT_1 \ln 2 \quad \therefore T_1 = 2T_2$$

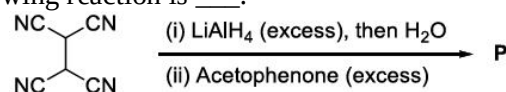
$$\Delta G_2^\circ - \Delta G_1^\circ = RT_2 \ln 2 + 2RT_2 \ln 2$$

$$= 3RT_2 \ln 2 = RT_2 \ln 8$$

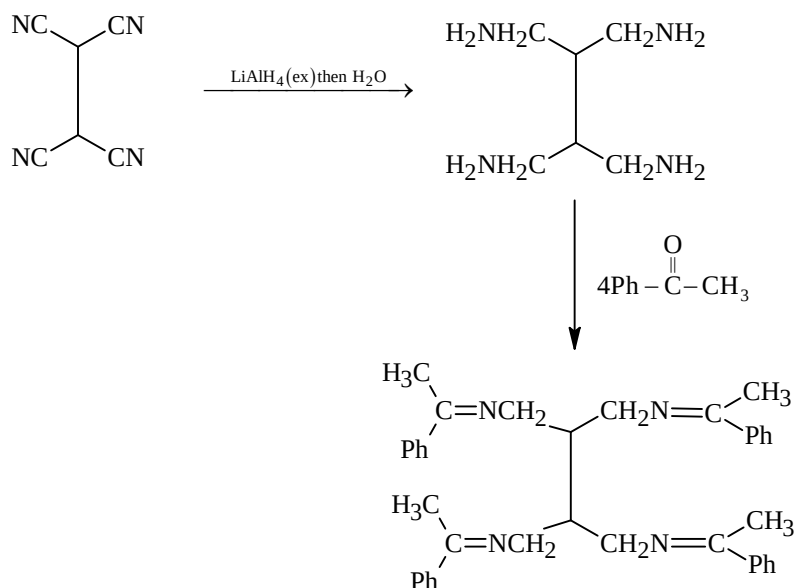
$$RT_2 \ln 8 = RT_2 \ln X$$

So, $X = 8$

- Q.13 The total number of sp^2 hybridised carbon atoms in the major product **P** (a non-heterocyclic compound) of the following reaction is ____.



Sol. (28)

**SECTION 4 (Maximum Marks: 12)**

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

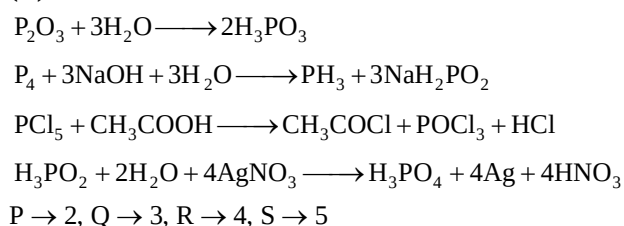
Full Marks	:	+3 ONLY if the option corresponding to the correct combination is chosen;
Zero Marks	:	0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-1 In all other cases.

Q.14 Match the reactions (in the given stoichiometry of the reactants) in List-I with one of their products given in List-II and choose the correct option.

List-I	List-II
(P) $\text{P}_2\text{O}_3 + 3\text{H}_2\text{O} \rightarrow$	(1) $\text{P}(\text{O})(\text{OCH}_3)\text{Cl}_2$
(Q) $\text{P}_4 + 3\text{NaOH} + 3\text{H}_2\text{O} \rightarrow$	(2) H_3PO_3
(R) $\text{PCl}_5 + \text{CH}_3\text{COOH} \rightarrow$	(3) PH_3
(S) $\text{H}_3\text{PO}_2 + 2\text{H}_2\text{O} + 4\text{AgNO}_3 \rightarrow$	(4) POCl_3
	(5) H_3PO_4

- (A) $\text{P} \rightarrow 2; \text{Q} \rightarrow 3; \text{R} \rightarrow 1; \text{S} \rightarrow 5$
 (B) $\text{P} \rightarrow 3; \text{Q} \rightarrow 5; \text{R} \rightarrow 4; \text{S} \rightarrow 2$
 (C) $\text{P} \rightarrow 5; \text{Q} \rightarrow 2; \text{R} \rightarrow 1; \text{S} \rightarrow 3$
 (D) $\text{P} \rightarrow 2; \text{Q} \rightarrow 3; \text{R} \rightarrow 4; \text{S} \rightarrow 5$

Sol.



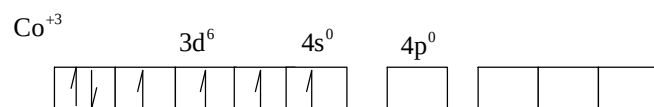
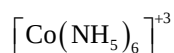
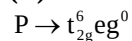
- Q.15 Match the electronic configurations in List-I with appropriate metal complex ions in List-II and choose the correct option.

[Atomic Number: Fe = 26, Mn = 25, Co = 27]

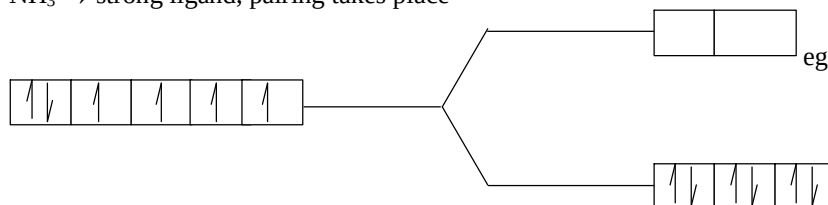
List-I	List-II
(P) $t_{2g}^6 e_g^0$	(1) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
(Q) $t_{2g}^3 e_g^2$	(2) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$
(R) $e^2 t_2^3$	(3) $[\text{Co}(\text{NH}_3)_6]^{3+}$
(S) $t_{2g}^4 e_g^2$	(4) $[\text{FeCl}_4]^-$
	(5) $[\text{CoCl}_4]^{2-}$

- (A) P $\rightarrow$ 1; Q $\rightarrow$ 4; R $\rightarrow$ 2; S $\rightarrow$ 3
 (B) P $\rightarrow$ 1; Q $\rightarrow$ 2; R $\rightarrow$ 4; S $\rightarrow$ 5
 (C) P $\rightarrow$ 3; Q $\rightarrow$ 2; R $\rightarrow$ 5; S $\rightarrow$ 1
 (D) P $\rightarrow$ 3; Q $\rightarrow$ 2; R $\rightarrow$ 4; S $\rightarrow$ 1

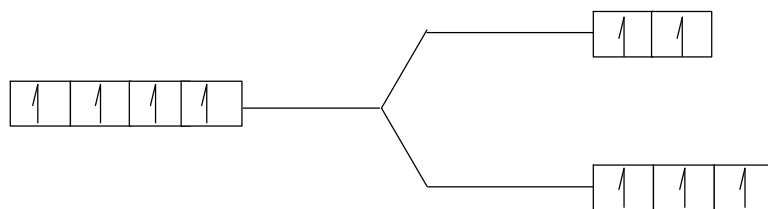
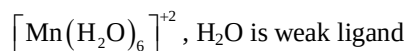
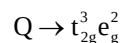
Sol. (D)



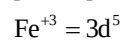
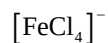
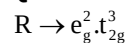
NH₃ $\rightarrow$ strong ligand, pairing takes place

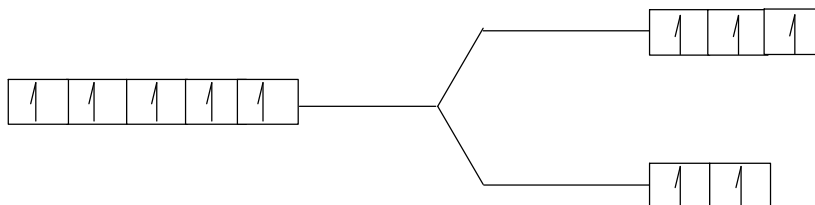


P - 3



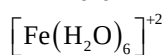
Q $\rightarrow$ 2



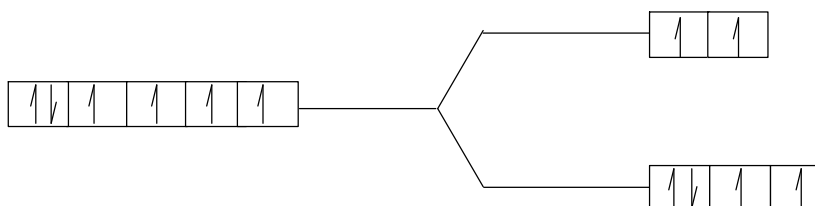


$$R \rightarrow 4$$

$$S \rightarrow t_{2g}^4 e_g^2$$

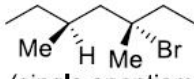


$$\text{Fe}^{+2} = 3d^6$$



$$S \rightarrow 1$$

Q.16 Match the reactions in List-I with the features of their products in List-II and choose the correct option.

List-I	List-II
(P) (-)-1-Bromo-2-ethylpentane (single enantiomer) $\xrightarrow[\text{S}_\text{N}2 \text{ reaction}]{\text{aq. NaOH}}$	(1) Inversion of configuration
(Q) (-)-2-Bromopentane (single enantiomer) $\xrightarrow[\text{S}_\text{N}2 \text{ reaction}]{\text{aq. NaOH}}$	(2) Retention of configuration
(R) (-)-3-Bromo-3-methylhexane (single enantiomer) $\xrightarrow[\text{S}_\text{N}1 \text{ reaction}]{\text{aq. NaOH}}$	(3) Mixture of enantiomers
(S)  (single enantiomer) $\xrightarrow[\text{S}_\text{N}1 \text{ reaction}]{\text{aq. NaOH}}$	(4) Mixture of structural isomers
	(5) Mixture of diastereomers

(A) $P \rightarrow 1$; $Q \rightarrow 2$; $R \rightarrow 5$; $S \rightarrow 3$

(B) $P \rightarrow 2$; $Q \rightarrow 1$; $R \rightarrow 3$; $S \rightarrow 5$

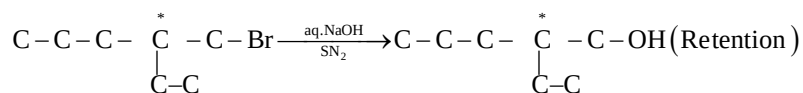
(C) $P \rightarrow 1$; $Q \rightarrow 2$; $R \rightarrow 5$; $S \rightarrow 4$

(D) $P \rightarrow 2$; $Q \rightarrow 4$; $R \rightarrow 3$; $S \rightarrow 5$

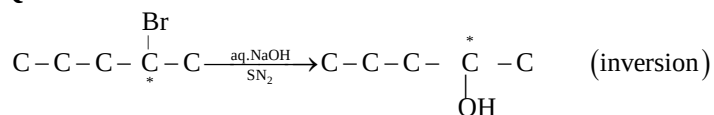
Sol.

(B)

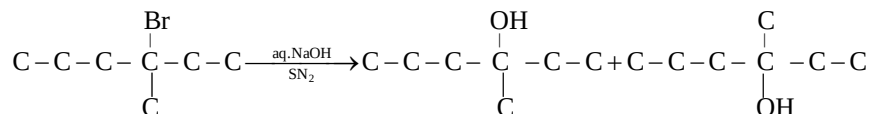
$P \rightarrow 2$



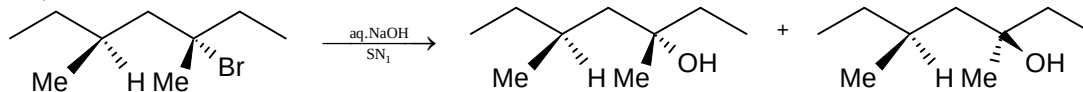
Q → 1



R → 3



S → 5



Q.17 The major products obtained from the reactions in List-II are the reactants for the named reactions mentioned in List-I. Match List-I with List-II and choose the correct option.

List-I	List-II
(P) Etard reaction	(1) Acetophenone $\xrightarrow{\text{Zn-Hg, HCl}}$
(Q) Gattermann reaction	(2) Toluene $\xrightarrow[\text{(ii) SOCl}_2]{\text{(i) KMnO}_4, \text{KOH}, \Delta}$
(R) Gattermann-Koch reaction	(3) Benzene $\xrightarrow[\text{anhyd. AlCl}_3]{\text{CH}_3\text{Cl}}$
(S) Rosenmund reduction	(4) Aniline $\xrightarrow[273-278 \text{ K}]{\text{NaNO}_2/\text{HCl}}$
	(5) Phenol $\xrightarrow{\text{Zn}, \Delta}$

(A) P → 2; Q → 4; R → 1; S → 3

(B) P → 1; Q → 3; R → 5; S → 2

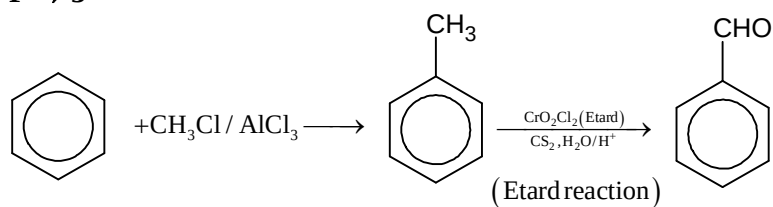
(C) P → 3; Q → 2; R → 1; S → 4

(D) P → 3; Q → 4; R → 5; S → 2

17.

(D)

P → 3



Q → 4

